



Section A

Answer all the questions in this section.

Question 1

Repeat droughts may cause permanent damage to forests

By: [Alex Whiting](#)

ROME, Aug 9 (Thomson Reuters Foundation) - **Trees and their environment need as long as two years to recover from drought in some places, and if a second dry spell hits before then, it may cause permanent damage to the landscape,** researchers said on Wednesday.

With **climate change expected to bring more frequent and intense droughts**, the implications for areas that do not have time to bounce back fully could be severe, the researchers said in a paper to be published in *Nature* journal this week. "That could have a double whammy effect," said co-author William Anderegg, assistant professor of biology at the University of Utah. "A second drought could be harder on an ecosystem and have the potential to push it off a cliff."

In practice, **that means affected areas could eventually turn from lush forest to a land of grass and shrubs.** Boreal forests in northern parts of Europe, Russia and Canada can take up to two years to recover from drought, partly because they do not have a wide variety of plants, Anderegg told the Thomson Reuters Foundation. Forests in the tropics of South America and Southeast Asia have also taken the same amount of time to rebound.

"That's worrisome because those regions store the largest chunks of carbon in ecosystems across the globe," Anderegg said. Forests help tackle climate change by sucking carbon out of the air, reducing levels of planet-warming carbon dioxide, the main greenhouse gas. But **when trees die, most of the carbon they have absorbed is released back into the atmosphere.**

The Amazon rainforest suffered a double drought in the first decade of this century when dry spells, both of a once-in-a-100-years severity, hit the region. "Satellites showed that forests hadn't recovered from the 2005 drought by the time the 2010 drought struck," Anderegg said.

Adapted from a Reuters Article

(a) Discuss how anthropogenic climate change has affected the growth of forests in the world. [4]

- Reduction in water content causes stomata to close which limits gaseous exchange thus affecting the availability of CO₂ for photosynthesis. (Negative Impact on Growth)
- Amount of RUBP carboxylase also drops significantly in low water content, leading to a decrease in CO₂ fixation in the Calvin cycle. (Negative Impact on Growth)
- Turgor pressure of cells and hence plant cell expansion is also slowed down, leading to retardation of plant growth. (Negative Impact on Growth)

- Increasing temperatures up to the optimal can increase plant growth and yield and rates of photosynthesis and respiration as a result of increased enzymatic activity. (Positive Impact on Growth)
- However, beyond the optimum, enzymes are denatured and plant death will result. (Negative Impact on Growth)

Any 4

(b) The article talks about how forests may not be able to recover if a second drought soon after a first. Suggest how forests may recover from a drought and explain why a second drought soon after a first impedes its ability to recover. [4]

- Forests recover from a drought by increasing rates of growth
- Forests recover from a drought by reproducing to increase their numbers
- All these take a significant length of time/ time for plants to grow; time for new plants to be formed and then grow
- A second drought will return the forest to a state before its recovery impeding its recovery/ reverse growth and reproductive gains

(c) The article talks about forests storing carbon as shown by the following quote *"That's worrisome because those regions store the largest chunks of carbon in ecosystems across the globe," Anderegg said.* Define the role of forests in storing carbon and explain briefly how they are able to do this. [4]

- They act as carbon sinks
- A carbon sink refers to a natural or artificial reservoir that accumulates and stores some carbon-containing chemical compound for an indefinite period.
- The process by which carbon sinks remove carbon dioxide from the atmosphere is known as carbon sequestration.
- This is done by the removal of carbon dioxide from the air during photosynthesis
- and the storage of this carbon dioxide in the form of biomolecules like carbohydrates, proteins and lipids

Any 4

Climate change does not just affect our forests, it also affects our farms. The following article elaborates on this.

Aug. 4 (UPI) -- Scientists from Lancaster University suggest major changes in agricultural practices are needed to offset increases in nutrient losses due to climate change. The study, shows that phosphorus losses will continue to increase due to climate change unless major changes in agricultural practices are made.

"Although farmers are already doing what they can to prevent these losses, the currently adopted measures are not likely to be enough to offset the increase expected under climate change. This paper should alert policy makers and government to the help and support that farmers will need to achieve the scale of agricultural change that may be necessary to keep up with the increase in pollution due to climate change."

Although phosphorus and nitrogen are essential to crop and animal growth, too much of it can cause algae blooms in rivers and lakes.

(d) Phosphorus and nitrogen are essential nutrients for plant growth and are typically found in large concentrations within the fertilizers used on farms. Suggest how climate change can lead to phosphorus and nitrogen from fertilizers ending up in rivers and water bodies. [2]

- **Heavy rainfall**
- **Washes the fertilizer off the land and into nearby rivers and water bodies.**

(e) The article talks about nitrogen and phosphorus causing algal blooms in rivers. Using a specific example of a plant or animal, discuss the possible impact of such blooms on the natural ecosystem in these rivers. [4]

- **Specific named typed of organism [eg. Fish, plankton, aquatic insects]**
- **Death or migration of the organism**
- **With explanation of how algal bloom lead to death or migration eg. Lack of oxygen in the water**
- **Elaboration of impact of loss of organism on the ecosystem**

(f) The article suggests that farmers need to change their agricultural practices to limit the impact of phosphorus used in fertilizer on nearby aquatic bodies. Describe two changes they could make and suggest why it may be difficult for them to make such changes. [3]

- **Use of plant strains that don't require as much fertilizer/crop rotation to minimize dependence on fertilizer/farming away from water bodies/AVP [any one properly described]**
- **Lank of funding/Lack of skill/Enculturation/AVP [Any one properly linked to an earlier change]**

Another aquatic ecosystem affected by climate change are coral reefs. The following graph, **Figure 1**, shows how sea water temperatures have varied over the last century or so at the Great Barrier Reef. 0°C is considered the average sea water temperature.

Rising sea level temperatures have been suggested as a reason for more frequent coral bleaching events. **Figure 2** shows the trend for coral bleaching events globally since 1980.

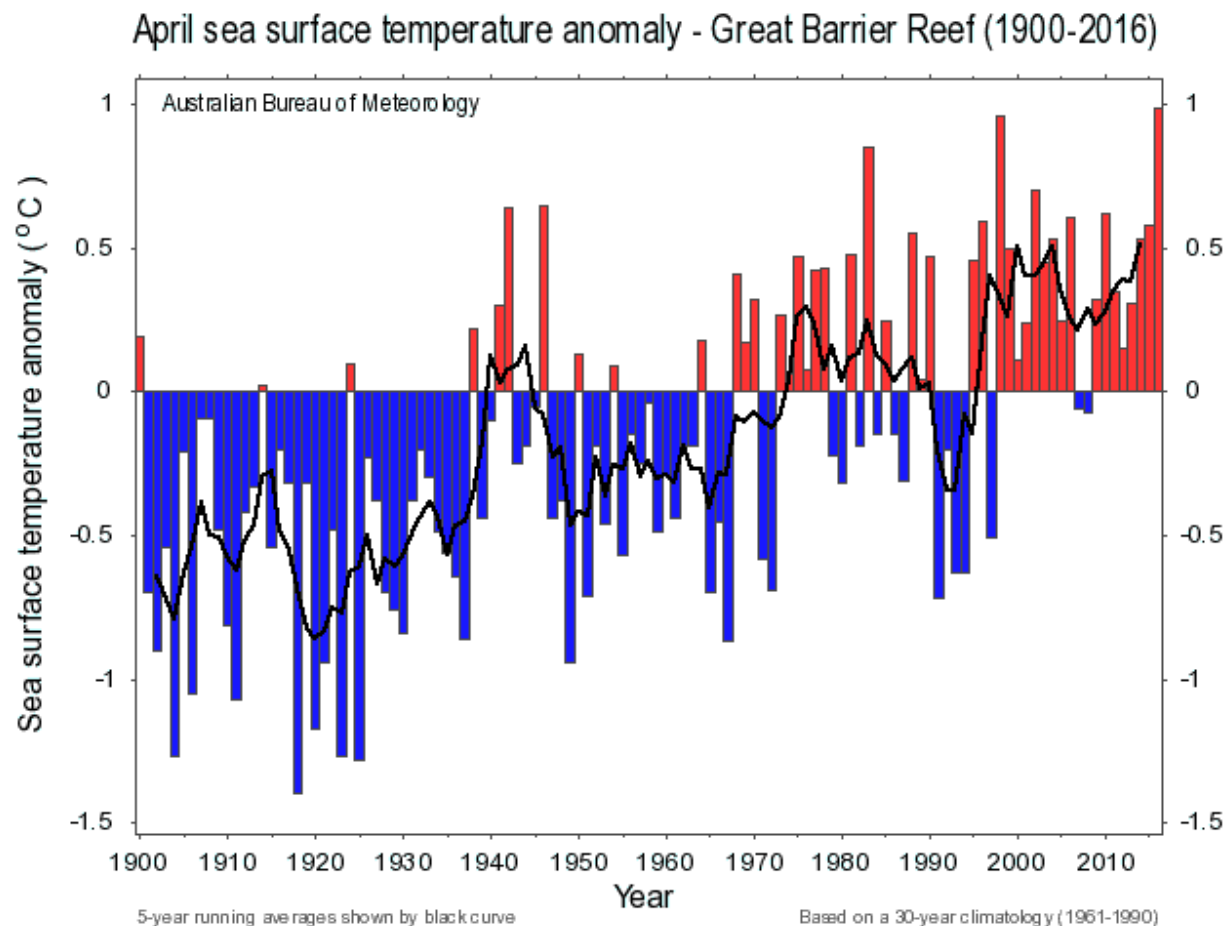


Figure 1

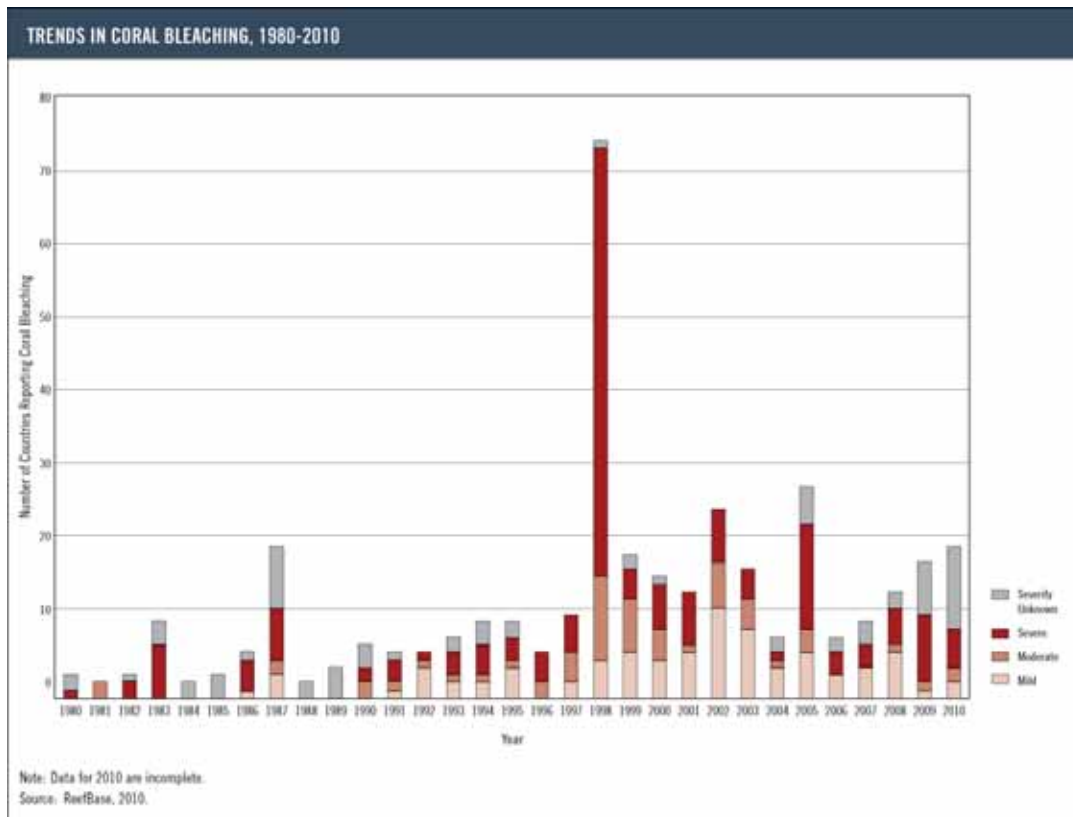


Figure 2

(g) With reference to **Figures 1 & 2**, discuss if the data provides evidence that the more frequent bleaching events seen in coral reef ecosystems over the last 15 years are due to rising water temperatures. [6]

- **Yes there is evidence.**
- **Most/More years above the average as compared with past 100 years. (Quote data)**
- **Peaks of high temperatures getting higher compared with past 100 years. (Quote data)**
- **Above average more frequent in the last 20 years compared with past 100 years. (Quote data)**
- **There have also been higher numbers of countries reporting coral bleaching events in the last 15 years. (Quote Data)**
- **There have also been higher number of countries reporting moderate to severe events in the last 15 years. (Quote Data)**
- **Higher sea temperatures can thus be directly correlated to coral bleaching events.**

(h) Other than warming water temperatures, describe two other ways anthropogenic climate change has impacted coral reefs. [2]

- **Rising sea levels**
- **Acidification of oceans**

[Total: 29 marks]

Question 2

A method that has been used to sequence DNA is called Sanger Sequencing, named after the scientist that invented the process. Sequencing DNA essentially allows the sequence of bases on a DNA strand to be read. During Sanger sequencing, DNA polymerases copy single-stranded DNA templates by adding nucleotides to a growing chain (**extension product**). Chain elongation occurs at the 3' end of a primer, an oligonucleotide that anneals to the template. The extension product grows by the formation of a phosphodiester bridge between the 3'-hydroxyl group on the primer and the 5'-phosphate group of the incoming deoxynucleotide.

DNA polymerases can also incorporate analogues of nucleotide bases. The dideoxy method of DNA sequencing developed by Sanger et al. 1977 takes advantage of this characteristic by using 2',3'-dideoxynucleotides (ddNTPs) as substrates. When dideoxynucleotides are incorporated at the 3' end of the growing chain, chain elongation is terminated selectively at A, C, G, or T. This leads to the production of a DNA fragment with the ddNTP at its end.

The ddNTPs are usually labelled using fluorescent tags. Many fragments of the original DNA strand terminated in the manner shown in **Figure 3** are created using this processes. When all these fragments are separated on a gel using a form of gel electrophoresis, the sequence of bases can be read by reading the labelled ddNTPs from the shortest to the largest fragment.

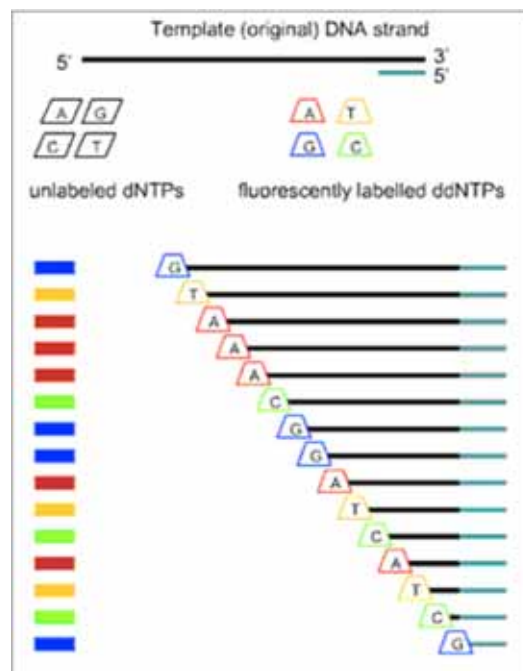


Figure 3

(a) Suggest how the labelling of the ddNTPs allows for the identification of the base that normally occupies that position. [2]

- **ddNTPs will only occupy a position at the very end of the strand.**
- **Different ddNTPs with different bases are labelled with a different fluorescent tag.**
- **Normal nucleotides (non ddNTPs) are unlabelled and would not show up under fluorescent treatment.**

(b) Describe how the extension product (line 4) is created and explain the role of the template DNA strand to this process. [5]

- **DNA polymerase extends the 3' end of the growing/daughter strand**
- **By adding deoxyribonucleotides via complementary base pairing with the bases on the template strand**
- **A:T, C:G**
- **Via hydrogen bonding**
- **Extension product (line 4) was created when DNA polymerase reads a 'T' on template strand, it adds an adenine ddNTP to the daughter strand**
- **Template strand thus carries the code for the sequence on the growing strand**
- **Template strand also ensures that the product is stable due to the hydrogen bonding with the template strand**

Any 5

(c) Explain how, when separated using gel electrophoresis, the strands produced by Sanger Sequencing end up in different positions on the gel. [3]

- **Strands are of different lengths**
- **Gel electrophoresis separates strands based in their lengths. Shorter fragments will move longer distances on the gel while longer fragments will move shorter distances**
- **Due to the resistance introduced by the gel that the fragments need to overcome (idea)**

The following **Figure 4** shows an example of dideoxynucleotide (ddNTP).

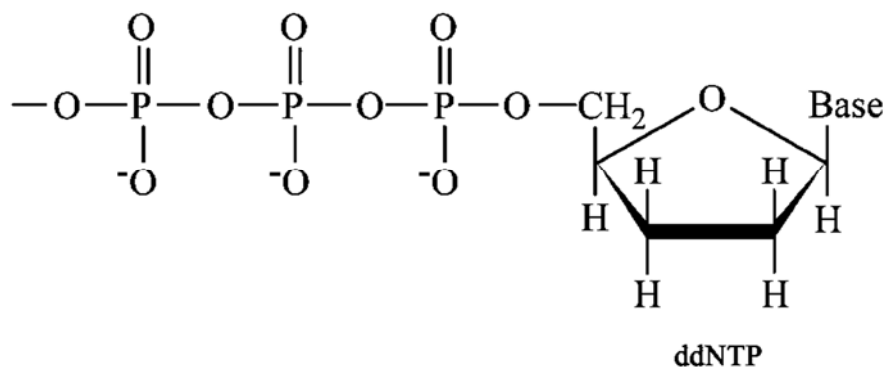


Figure 4

(d) Explain why dideoxynucleotides, such as the one shown in Figure 4, leads to the formation of fragmented DNA. [3]

- **No 3'-OH group**
- **DNA polymerase requires a 3'-OH group to extend a DNA strand.**
- **DNA polymerase cannot attach and extend growing strand, hence producing fragments of DNA.**

- (e) Methods such as Sanger Sequencing has enabled the sequencing of the human genome. This has opened up the possibility of detecting specific alleles in human genomes. Discuss the possible advantages of this and its possible ethical ramifications. [4]

Possible Advantages

- **Improved diagnosis of disease**
 - **Earlier detection of genetic predispositions to disease**
 - **Drug design**
 - **Gene therapy and control systems for drugs**
 - **Identify potential suspects whose DNA may match evidence left at crime scenes**
 - **Exonerate persons wrongly accused of crimes**
 - **Identify crime and catastrophe victims**
 - **Establish paternity and other family relationships**
 - **Match organ donors with recipients in transplant programs**
 - **Study evolution through mutations in lineages**
- Any 2 with brief elaboration

Possible Ethical Ramifications

- **Genetic testing - psychological impact and stigmatization**
 - **Genetic Testing - fairness in the use of genetic information and privacy and confidentiality**
 - **Reproductive issues - Prenatal testing / Reducing the number of deformities**
 - **Preimplantation testing - Gender selection**
 - **Commercialization - questions of the ownership of tissue and tissue derived products, patents, copyrights, and accessibility of data and materials**
 - **Conceptual and philosophical implications - free will vs genetic determinism / Do people's genes make them behave in a particular way? Can people always control their behavior?**
- Any 2 with brief elaboration

Human genomes include the DNA found in the mitochondria. Similar to the nuclear genome, the mitochondrial genome is made up of double-stranded DNA, and it encodes genes. However, the mitochondrial genome differs from the nuclear genome in several ways

- (f) Suggest two ways in which mitochondrial DNA may differ from the nuclear genome. [2]

- **The mitochondrial genome is circular, whereas the nuclear genome is linear.**
- **The mitochondrial genome is smaller/ less DNA base pairs, whereas the nuclear genome is larger/more DNA base pairs.**
- **The mitochondrial genome contains less genes.**

- (g) With reference to specific genes, explain the role of mitochondrial DNA. [2]

- **It carries genes that code for proteins used in the mitochondria.**
- **Examples include tRNA gene, rRNAs genes, genes of oxidative phosphorylation carrier proteins (any two logical genes stated)**

[Total: 21 marks]

Section B (25 Marks)

Answer **only one** question

Write your answers in the space provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 3

- (a) All organisms need to replicate and transcribe/translate their DNA in the process of growth and development. Even viruses need to do so. Using the example of a dengue virus, compare and contrast the replication and protein synthesis process in a virus with the similar processes in a typical eukaryotic cell. [13]

Similarities

- Both use polymerases in replication and transcription
- Both use ribosomes in protein synthesis
- Both use amino acids as raw material for protein synthesis
- Both produce polypeptides in protein synthesis
- Enzymes used in protein synthesis include peptidyl transferase and aminoacyl t-RNA synthetase for both.

Differences

Criteria	Dengue Virus	Eukaryote
Polymerase for Replication	RNA Dependent RNA polymerase	DNA polymerase
Transcription	Not required as genome is positive RNA	Carried out by DNA Dependent RNA Polymerase
Replication	Two rounds of replication required to achieve product (positive strand RNA)	A single round of semi-conservative replication is sufficient
Template for Replication/Product of Replication	RNA/RNA	DNA/DNA
Raw material for replication	RNA nucleotides	DNA nucleotides
Source of replication and protein synthesis machinery	Host Cell	Own Cell
Unwinding of template	No unwinding as template for both Replication and Transcription is single stranded	Unwinding/separation of DNA template for replication carried out by helicase while for transcription carried out by RNA polymerase.
Post transcriptional modification	Absent	Present

Max 12

QWC – Criterion-based comparison (1 mark)

(b) Describe how viruses such as the influenza virus is able to create genetic variation and explain how this makes it difficult for us to eradicate harmful viruses with modern medicine. [12]

- Via antigenic drift
- Antigenic drift is due to spontaneous mutations (e.g. point mutation) of the viral gene encoding haemagglutinin glycoprotein (antigen)
- Such mutations result in the production of new hemagglutinin proteins that are of a different conformation.
- Antibodies made against the original influenza virus can no longer recognize and bind to the new haemagglutinins antigens, thus the virus can evade host immune defenses.
- Mutations in influenza occur frequently because the viral RNA polymerase has no proofreading mechanism, providing a strong source of mutations.

Max 4

- Via antigenic shift
- Antigenic shift occurs when 2 different strains of influenza virus infect a single host cell at the same time
- When two different strains of influenza infect the same cell simultaneously, their protein capsids and lipid envelopes are removed, exposing their RNA genome.
- The RNA genome segments reassort and are randomly incorporated during assembly of new virus, thus forming a novel subtype of virus that has a mixture of glycoprotein antigens of 2 original strains.
- For example, H3N2 and H5N1 can form H5N2.
- Appropriate diagram

Max 4

- Many medicines target the glycoproteins on the influenza virus envelopes.
- To prevent adsorption and penetration.
- By physically blocking the receptor sites on the host cell
- Or interfering with the glycoproteins in the virus
- When the glycoproteins change due to antigenic shift and drift, the medicines will no longer work
- As they are specific to the glycoproteins they were designed to target.

Max 4

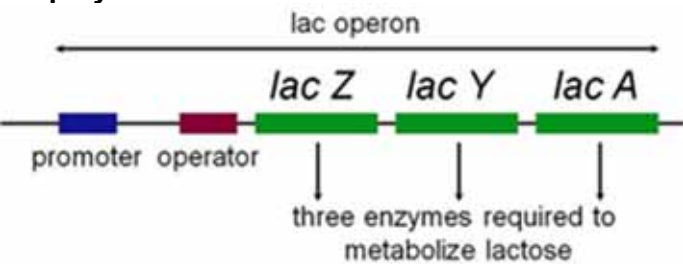
Question 4

- (a) All organisms need to control the expression of their DNA in the process of growth and development. Even bacteria need to do so. Using the example of *E. coli*, compare and contrast the control of gene expression in a bacteria with the similar processes in a typical eukaryotic cell. [13]

Similarity

- Expression of gene involves transcription and translation for both.

Differences

Control of Gene Expression		
Feature	<i>E. coli</i> Gene Expression	Eukaryotic Gene Expression
Genome Level	• Chromatin modification e.g. histone acetylation cannot occur as prokaryotic DNA is not associated with histones.	• Chromatin modification e.g. histone acetylation can occur, resulting in conversion between euchromatin and heterochromatin. ◦ Eukaryotic DNA is associated with histones (and other proteins).
	• DNA sequences (promoters and operators) serve as on/off switches.	• Degree of condensation of chromatin serves as major on/off switches for gene expression.
	• Gene amplification does not occur	• Gene amplification occurs to increase number of copies of the gene of interest.
Transcription Level	• Related genes are organised in an operon, under the control of a single promoter. • These genes are transcribed together to give rise to a polycistronic mRNA. 	• Genes are not organised into operons. • Each gene has its own promoter and gives rise to a monocistronic mRNA.

	• Few control elements, which are usually located close to the promoter and genes under its control. - o E.g. operator is located close to the promoter of genes.	• Many control elements that can be located proximally or distally upstream/ downstream of a gene. - o E.g. proximal and distal enhancers and silencers.
	• Only one RNA polymerase involved. All RNAs are synthesised by the same RNA polymerase.	• Five different RNA polymerases present. - o Three main types of RNA (mRNA, tRNA and rRNA), synthesised by three different RNA polymerase.
	• General transcription factors not required as RNA polymerase can directly recognize and bind to pribnow box of prokaryotic promoters. - o Occurs with the aid of sigma factors which reduces the affinity of RNA polymerase for non-specific DNA sequences, and increases its affinity for the promoter. - o Prokaryotic RNA polymerase associates with the sigma factor to form RNA polymerase holoenzyme.	• General transcription factors required. - o General transcription factors and RNA polymerase II assemble at the TATA box of eukaryotic promoters to form the transcription initiation complex.
Post-transcriptional Level	• Post-transcriptional modifications do not occur. Primary transcripts are the actual mRNA.	• Primary transcripts (pre-mRNA) undergo processing to produce mature mRNA; - o Addition of 5' cap - o RNA splicing - o Addition of 3' poly (A) tail
	• Lower stability of mRNA transcript - o Degradation occurs within seconds to minutes of transcription. - o Allows prokaryotes to respond rapidly to environmental changes.	• Higher stability of mRNA transcript - o Degradation after minutes or days following transcription - o Degradation controlled by length of poly (A) tail → the

		longer the poly(A) tail, the later the mRNA is degraded.
Translational Level	Translation is often coupled to transcription. Both processes occur simultaneously.	- Translation is not coupled to transcription. - mRNA must move from the nucleoplasm, across the nuclear envelope, via a nuclear pore to the cytoplasm for translation to occur. - RNA transcript is not free to associate with ribosomal subunits prior to completion of transcription.
	Control at this level is unlikely, due to simultaneous transcription and translation.	Control can occur at pre-translational level, when regulatory proteins bind at the 5' UTR or the 3' poly (A) tail of mature mRNA. This prevents binding with the small ribosomal subunit and the assembly of the translation initiation complex.
	mRNAs are polycistronic and have multiple start codons, allowing for the direct synthesis of several different polypeptides.	mRNAs are monocistronic and have only one start codon, allowing for the synthesis of only one kind of polypeptide.
Post-translational Level	Post-translational modification does not occur	- Post-translational modification can occur in the form of; - Chemical modification (glycosylation and phosphorylation) - Proteolytic Cleavage and Activation - Degradation (ubiquitination)

Max 12

QWC – Criterion-based comparison (1 mark)

- (b) Describe how bacterial cells are able to create genetic variation and explain how this makes it difficult for us to eradicate harmful bacteria with modern medicine. [12]

- Transformation
- Transformation is the process by which a naked, foreign DNA molecule is taken up from the surrounding external environment
- and integrated into the bacterium's genome, via homologous recombination
- thus resulting in the change of recipient bacterial cell's genotype.
- The resultant transformant cell will now express genes from this new segment it has received, and pass them on to all subsequent daughter cells by binary fission.

Max 3

- Transduction
- Transduction is the process by which DNA is transferred from one bacterial cell (donor) to another (recipient) by bacteriophages,
- and is later integrated into the bacterium's genome of the recipient cell via homologous recombination.
- During the assembly stage, phage genome is randomly packaged within the phage capsid to form the mature phage particles. But occasionally, a small piece of the host cell's degraded DNA is packaged within the phage capsid in place of the phage genome.
- This 'defective' phage can attach to another recipient bacterium and inject the piece of bacterial DNA acquired from the first cell

Max 3

- Conjugation
- Conjugation is the process by which bacterial cells make direct contact with each other and DNA is directly transferred from one donor cell to the another recipient cell,
- The DNA donor uses appendages called sex pili for the transfer
- Sex pili attach the F⁺ donor to the F⁻ recipient cell.
- The sex pili facilitate the direct DNA transfer between donor and recipient cells by forming a temporary cytoplasmic bridge (also known as conjugation tube or mating bridge)
- the F plasmid replicates within the donor cell, and only one strand of the plasmid is transferred to the recipient through the conjugation tube joining the cells.

Max 3

- Modern medicine uses antibiotics to target bacteria.
- Antibiotics do not work on bacteria carrying genes for resistance to that antibiotic.
- Such antibiotic resistance genes can be transferred between bacterial cells via transformation, transduction or conjugation.
- This increases the number of resistant bacterial cells making it difficult to eradicate the bacteria.

Max 3